Measurement as Commitment

The quantum measurement process as the contractive half of one operator on D_IV^5

Draft, 2026-08-04 (finished 2026-08-22, R50: §2a commit-operator dig absorbed as a permanent subsection; odds-Derived reconciled per Elie's verified sort-weighting; spine aligned to the corrected forcing claim — n_C=5 is one measured input, N_c=3 the ruler, W10). Lyra, from the Koons measurement conversation (Aug 3–4). Honestly tiered throughout — every claim marked Derived / Identified / Posit / Open. This paper + its sober companion "The Axioms of Quantum Mechanics from D_IV^5" (10/10 Dirac–von Neumann axioms, zero posits) are THE named artifact for internal-rubric criterion C (the commitment ontology). Cal hostile-read + Keeper pass before any external send; nothing sent without Casey's direction. The one word we do not write is "solved."

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1. The open

Quantum mechanics has one axiom that has never been derived and one that has never been comfortable, and they are usually stated as different problems. The uncomfortable one is the Born rule: probabilities are the *squares* of amplitudes — why squares? The underived one is measurement: a smoothly-evolving superposition somehow becomes a single definite outcome — why, and how, and which?

This paper's claim is that on one geometry these are not two problems but two readings of one object. The Standard-Model framework Bubble Spacetime Theory places physics on a single bounded symmetric domain, D_IV^5 = SO(5,2)/[SO(5)×SO(2)], whose states live in the Bergman/Hardy space H^2(D_IV^5) and whose dynamics are generated by one operator — the Casimir H_B of the maximal compact subgroup. We show that measurement is the *contractive half* of that one generator, that the Born rule is its forced measure, and that the single outcome is a boundary datum of exactly the kind every physical theory — including general relativity — takes as given. What remains genuinely open is one dynamical computation, stated precisely, and one interpretive posit, stated as a posit.

> *For a general reader:* imagine the world runs on a shape, and a single dial on that shape. Turn the dial one way and things flow smoothly and reversibly — that is ordinary quantum evolution. Turn it the other way and things settle, once, irreversibly — that is a measurement. Same dial. The odds on the settling are baked into the shape. This paper is about that dial.

2. One operator, two faces

The generator H_B has two exponentials, and they are the two things quantum mechanics does.

- The unitary face, exp(i τ H_B/$\hbar$) — reversible. This is Schrödinger evolution: superpositions form, correlations build, interference lives here. Call it *absorb*. (Derived: H_B is the Casimir; unitary evolution is its standard action.)
- The contractive face, exp($-\tau$ H_B/$\hbar$) — the heat semigroup, irreversible. Its positivity is the arrow of time, which BST already derives. This is the *commit*: the collapse toward a definite state. (Derived: the contractive semigroup and its irreversibility.)

The measurement problem, in its standard form, is a problem *only* for a theory with the unitary face alone — with no physical collapse, there is no single outcome, which is why many-worlds keeps every branch and Copenhagen posits an external cut. BST is not such a theory: the same generator that turns smoothly also settles. Collapse is not added; it is the operator's other exponential.

2a. The commit's own structure: write, and check

Section 2 names the commit as the contractive face of H_B. The forced-object dig (2026-08-16, K1607/K1609) sharpened *what that commit is internally*: it is not a structureless settling but one graded operator,

C = P_record ⊕ P_encode,

the geometry's own Peirce decomposition of a primitive idempotent — the commit read one way. Its two summands are the two things an integrity-checked irreversible record must do:

- P_record — the WRITE. A rank-1 projection onto the *arrow line*: an irreversible binary record laid on the Shilov boundary. "This happened," undoable — the same one-sidedness that is the arrow of time (Section 2). This is where the domain's rank = 2 enters, and it enters from the *structure of a binary outcome*: a commitment has two complementary outcomes — the idempotent e (written) and its orthogonal complement 1 − e (not-written) — and the Jordan rank is the number of such mutually orthogonal outcomes, which is exactly 2. Two outcomes, rank two — the first of BST's five integers fixed by what a binary decision *is*, not by the physics we want to recover (a single rank-1 projection cannot exhibit rank 2; the two of them do). (*Derived — verified; the external-forcing of rank=2 is the novel content.*)
- P_encode — the CHECK. The projection onto the confined 3-dimensional block V_{12}: a detection/erasure structure (minimum distance 2) that wraps the record so a later corruption is *flagged (detected), but not located* — at d_min=2 you detect that damage occurred, you do not locate it (locating is correcting, which needs d ≥ 3). The record is therefore self-validating — it carries its own damage report. (*Derived. One correction kept in-text, because it matters*

> *to a referee: the encode is detection/erasure, not error-*correction*, so the record is self-validating but not "unforgeable" — a public code forges trivially; the tamper-resistance is the irreversibility of the write, not the code. Earlier drafts that said "error-correction/unforgeable" are corrected here.*)

Neither stage is removable — which is exactly why the commit is *minimal*: drop the encode and the record can be silently corrupted; drop the write and there is nothing to encode. Both stages are therefore *necessary*. Whether they are jointly *sufficient* — whether a commit must also be able to repair a detected corruption rather than merely flag it — is a separate question this section does not settle: irredundancy ("neither stage can be dropped") does not establish exhaustiveness ("no further stage is required"). "Write the bit; check the bit" is the contractive commit of Section 2 seen from the inside. *(Corrigendum, Cal §606: an earlier draft said this "is the whole operation" — an exhaustiveness claim the irredundancy argument above does not reach.)*

The honest floor (stated as a floor, not hidden — amended 2026-08-24; original four-posit form retired, see the amendment note). The chain *binary commitment ? rank 2 ? isotropic complement ? spin factor ? type-IV (Lorentzian)* is theorem-grade, and it rests on two posits plus one measured input: P0 (the arena is a Euclidean Jordan algebra — an axiom), P1 (a commitment is one *binary* distinction — the cheapest axiom, which does the rank-2/type-IV work. *Separately*, the resulting type-IV domain is homogeneous *by construction*, so BST *cannot derive* a preferred frame — an inability result, conditional on the 5→4 descent posit, and *not* a prediction that none exists), and the RULER (the measured N_c = 3, the program's one declared input, which pins n_C = 5 among the survivors {5, 11, 13, ...}). *Amendment note (2026-08-24):* P2a (exclusivity) is DISCHARGED — named once, invoked nowhere; all three readings discharge, two by theorem; P1's binarity removes the situation in which it would have content; the discharge is not circular because orthogonality in the *definition* of rank is a defined relation, costing no posit (rubric Internal A, 2026-08-23). P2b (minimality) is STRUCK from the floor — named once, invoked nowhere; where selection actually occurs the selector is the ruler (v0.7, K1697: *"the tiebreaker that pins it uniquely is the measured N_c = 3"*); where minimality has content it is v0.7 Appendix A's *theorem*. Conditional attached: the strike holds *given the declared-ruler stance*; reopening zero-input uniqueness re-admits minimality as selector and inherits this draft's own verdict on it — *degenerate in every lane, so it does not by itself deliver n_C = 5* (Lyra R85b, second CI Keeper, 2026-08-24). So the operation forces the *type* of the world (Lorentzian, type-IV) mod P0+P1, with rank=2 externally forced; the *dimension* n_C=5 is pinned by the declared measured input. Never "commitment forces D_IV^5." (*Type forced mod P0+P1 = Derived-conditional; rank=2 = Derived (external); dimension = input-pinned (the ruler), declared not derived.*) Ratified floor (2026-08-24, Cal gate + Keeper pass): P0 (arena) + P1' (a crossing that amplifies — Section 2b) + the ruler. Binarity: theorem. Irreversibility (irr_thermo): theorem, conditional on the second law (statistical-counting import). Exclusivity: discharged. Minimality: struck. P1 itself is now a theorem of (P0 + P1').

2b. The commit event is a boundary crossing *(added 2026-08-24 — the ratified interface formalization)*

Three words, three objects, pinned at first use: commit_op = the graded operator $C = P\_record \oplus P\_encode$ of Section 2a · commit_ev = the crossing *event* at the interface, defined here · commit_rec = the written, amplified record. Section 2a gave the commit's anatomy; this section gives its *geometry*, and with it the floor's last two soft spots become theorems.

The interface is the cone boundary. The arena's symmetric cone $\Omega \subset J$ has a boundary that is two-sided by convexity alone — no idempotent needs naming: $J \setminus \partial\Omega$ has exactly two components (the interior is convex, hence connected; any exterior point rides the funnel $x(t) = (1-t)x - t \cdot 1$, whose spectrum shifts rigidly and stays strictly outside — an exact three-line argument, valid at every rank, so the two-sidedness cannot be rank-2 in disguise). This is the substrate/continuum interface of the physical picture, formalized on the cone side.

P1' (the posit, ratified form): *a commitment (commit_ev) is a crossing of $\partial\Omega$ whose which-room datum is represented — amplified into redundant copies — in the container.* Both clauses are the physical picture (the crossing of the 2D/3D interface; the push paying for the representation to be written into the container), both stated before the mathematics needed them.

Binarity is then a theorem: the topological datum of a crossing is the ordered pair (room-from, room-to) — an element of a 2-set. One crossing, one bit.

Irreversibility is a priced theorem — with a subscript that matters. Two irreversibilities, two objects: irr_arrow, the absolute geometric arrow (the contractive-face positivity of Section 2), and irr_thermo, the cost of *reversing* a commit_ev: erasing every copy of the amplified commit_rec at $\geq n \cdot k\_BT \ln 2$ (the Landauer erasure bound — a statistical-counting import of the second law, not a second use of the arrow, so nothing is double-counted). A single-copy write is thermodynamically reversible (Bennett); *only the amplified record commits* — the amplification clause is load-bearing, not poetic. BST thus carries a two-tier arrow: geometric-absolute, and thermodynamic-priced. The commit rides the second.

3. Born and collapse are one measure

What are the odds on the settling? Here the geometry is decisive. On a bounded symmetric domain there is exactly one measure invariant under the domain's automorphisms — the Bergman/Faraut–Korányi measure; the ordinary "flat" (Lebesgue) measure is *not* invariant and is excluded. The Born rule, probability $= |\langle k|\psi\rangle|^2$, is precisely the density of that forced measure. (Derived: T754 — the invariant measure is unique and is the Bergman measure; the squared-amplitude rule is its density.)

So Born and collapse are not two axioms: the measure gives the weights, the contractive commit is the irreversible selection *from* those weights. One forced object underlies both.

> *For a general reader:* there is only one honest way to weigh possibilities on this shape — its own built-in measure — and "weigh by the square of the amplitude" turns out to *be* that one way. So the squares aren't a rule someone chose; they're the only self-consistent scale the shape allows.

4. The commit is a sharp sort, not a slow settling

A natural worry (and a real one — it was raised and had to be answered): if the commit were a *continuous* relaxation of the heat semigroup, it would drain toward the lowest-energy state and *bias* the odds — the heavier modes fade fastest, so the Born ratios would drift. We verified the drift explicitly.

The resolution, due to Casey Koons, is that the commit is not a continuous drain. It is a sharp sort against a ground reference: each amplitude either finds a finite-norm home — a real state, keeping exactly its share — or it diverges and leaves, and is not a state at all. Run *that* and the survivors keep their Bergman-measure ratios *exactly*; the divergent mode simply drops out, and the physical overlaps are untouched. We checked it: the continuous version biases (0.30:0.50:0.20 → 0.41:0.46:0.12); the sharp sort preserves (survivors stay 0.30:0.50 exactly).

And the ground reference is the domain's own: D_IV^5 sorts its modes into finite-norm unitary states (the holomorphic discrete series, the Wallach points) and negative-formal-degree divergent non-states (the reflected partners with $d(\text{genus}-\nu) = -d(\nu) < 0$; the "unphysical root" that recurs in the framework's own uniqueness analysis is one). Finite drains to a state; divergent runs to infinity and leaves. Casey's "wired to ground; amplitudes go to zero or infinity" is not a metaphor added on — it is the shape's own structure. And this is now forced, not conjectured (Elie, verified): the commit is the composition *Wallach-drop* ∘ *measure-keep* — the divergent modes are not states, so a commitment *cannot* land on them (the drop is forced by the geometry, not chosen), and the weighting among the surviving real states is the unique invariant measure (T754), which is Born. So the sharp sort, and the Born weights it preserves, are derived from the domain.

> *For a general reader:* the settling isn't a slow drift where the heavy things fade and skew the odds. It's a hard sort against a fixed reference — a thing either has a real, finite place to be, or it runs off to infinity and stops counting. The ones that stay keep their exact shares. That hard sort is the only version that keeps the odds honest, and the shape already has the reference built in.

5. Compute from now

Two further points close the account. First, there is no *fixed* ground for outcomes to pile into across many settlings, because each commit is a fresh start — the memoryless semigroup property: the committed state becomes the new initial condition and the reference re-grounds. Reality is the *sequence of commits*; between them is potentiality (the unitary "potential updates"). Casey's phrase is exact: the reality is the last one committed; compute from now. The sharp sort keeps the odds honest within a tick; the reset keeps them honest across ticks.

6. The single outcome is a boundary datum

This leaves the last question — *which* single outcome — and here we are careful, because this is where every account of quantum mechanics has always stopped, and honesty requires saying so. The contractive process explains why the world becomes definite and with what odds; it does not, and no theory does, *derive* which specific result occurs.

But this is not a hole peculiar to BST. It is the same kind of thing as an initial condition. General relativity does not derive its initial slice; it evolves whatever Cauchy data it is handed. The committed outcome is data of exactly that kind — the seed the next moment computes forward from. Asking "why this outcome" is asking "why this boundary datum," and no physical theory answers that; it takes it as given and evolves it. (Posit/boundary datum, by category — not a shortfall, a data-versus-dynamics distinction.)

And the residual randomness is a *feature*, not a gap: it is precisely *where quantum randomness comes from* in this picture. A deterministic universe would have no commit-randomness and no Born statistics; the stochastic commit *is* the physical origin of the odds. As Casey puts it, a commitment you could predict wouldn't be one.

7. What kind of theory this is

Cast in the standard taxonomy, BST is an objective-collapse theory — reality has real, physical collapse events, like GRW and CSL — but a *derived* one. GRW and CSL must *posit* the collapse, its rate, and its Born weights. BST forces all three from the geometry: - the collapse mechanism is the contractive face of the same H_B whose unitary face is Schrödinger (not an added noise term); - the Born weights are the forced Bergman measure (T754); - the rate is the substrate tick (T2405).

That is strictly more than the standard interpretations offer: a physical collapse where they have infinite branches or a hand-wave, with mechanism, weights, and rhythm all read off one shape rather than assumed.

8. Erasure and correlation

The two-face structure also explains the delayed-choice quantum eraser without extra machinery. Correlation is built by the *unitary* face and nothing collapses until the *contractive* commit — so erasing the correlation before the commit leaves nothing to collapse, and a correlation never committed never projects. This is forced by the domain carrying both faces of one H_B.

The quantitative Bell prediction is concrete and can-fail. BST predicts a sub-Tsirelson CHSH value: the deviation from the quantum bound is the exact BST-primary identity

$$\text{Tsirelson}^2 - S^2\text{\_BST} = \text{rank} / 2^\wedge\{\text{rank}^2\} = 1 / 2^\wedge\{N\_c\} = 1/8 \text{ (T2399)},$$

so S_BST = $\sqrt{}(8 - 1/8) \approx 2.806$, a 0.79% shortfall below Tsirelson's $2\sqrt{2} \approx 2.828$. This is a sharp, near-term falsifier: a high-precision CHSH measurement finding S $< 2\sqrt{2}$ supports BST at bounded level regardless of the exact value; finding S $= 2\sqrt{2}$ to that precision refutes it. *Tier: the sub-Tsirelson bound and the exact 1/2^{N_c} deviation are Identified (an exact algebraic identity, T2399); the specific value 2.806 = (N_c/rank)·√(g/rank) rides a BST-primary candidate form and is not yet derived from the substrate Hamiltonian — the operator-level CHSH identification (which K-type realizes the CHSH operator; Calibration #17) is the one remaining computation. It is non-blocking for this paper: the measurement process, the arrow, the odds, and the commit mechanism are already Derived (Section 9); the Bell magnitude is a separate falsifiable prediction, stated here at its honest tier.*

9. The honest ledger

piece	status
Schrödinger evolution = unitary face of H_B	Derived
arrow of time = contractive-face positivity	Derived
Born rule = the forced Bergman measure	Derived (T754)
becoming-definite = the contractive commit	Derived
the commit's internal structure = the graded operator C = P_record ⊕ P_encode (write ⊕ confined 3-encode, neither removable)	Derived (Peirce split of a primitive idempotent; K1607/K1609)
rank = 2 forced externally (binary write; Shannon alphabet size = Jordan rank)	Derived (external) — first BST integer forced from outside the geometry
binarity from the interface (commit_ev at $\partial\Omega$; $\pi_0 = 2$, ordered-pair datum — Section 2b)	Theorem (ratified 2026-08-24)
irreversibility irr_thermo (erasing the amplified record costs ≥ n·k_BT ln 2 — Section 2b)	Theorem, conditional on the second law (statistical-counting import; the two-tier arrow stated once, not double-counted)

piece	status
the edge-identity (fermion chamber edge = (n−1)/2, family-generic 4/4 at n = 3, 5, 7, 9)	Established-imported identification (pin LANDED 2026-08-25): the edge IS the known spinor unitarity bound — Minwalla, hep-th/9712074, eq. (2.56): "$\varepsilon_0 \geq (d-1)/2$ (spinor)," saturated by the free Dirac field (Sec. 2.5, verbatim: the saturating operator "obeys the Dirac equation"). BST content = the chamber structure REPRODUCES the bound family-generically; novelty down, solidity up, as pre-agreed
the encode is self-validating (detection/erasure d=2), not "unforgeable"	Derived (correction kept in-text)
commitment forces the *type* of spacetime (Lorentzian type-IV)	Derived-conditional (mod P0+P1); dimension n_C=5 stays an honest input
the odds survive the commit (sharp sort preserves Born)	Derived (Elie, verified: the commit = Wallach-drop ∘ measure-keep is forced — divergent modes are non-states so the drop is not chosen, and the surviving weights are the unique invariant measure T754 = Born)
erasure / Bell statistics	sub-Tsirelson prediction: S_BST ≈ 2.806, exact deviation $\text{Tsirelson}^2 - S^2$ = 1/2^{N_c} = 1/8 (T2399) — a can-fail Bell falsifier. Identified (the bound + the exact deviation); the specific value rides a candidate form, operator-level realization = the one open computation (non-blocking)
the single outcome	boundary datum (given, by category — as GR takes its initial data)

The claim, exactly as far as it goes: BST derives the *process* of measurement — the becoming-definite, the arrow, and the odds — as the contractive half of one operator, and leaves standing only the single-outcome boundary datum that every account of quantum mechanics leaves standing. (The odds are now derived, not conjectured: the commit is a forced Wallach-drop composed with the forced invariant measure; see Section 9.) This is not "measurement is solved." It is "everything about measurement that can be derived, is — and what remains is data, not dynamics." We hold the line there deliberately: the over-claim we refuse is any suggestion the specific outcome, or the fact of randomness itself, is explained away. It is physical, and irreducible, and that is correct.

Handoffs

- @Elie — closure computation (1) is DONE (verified, folded into Section 9): the commit = Wallach-drop ∘ measure-keep is forced, the survivors sit at $|\langle k|\psi\rangle|^2$ exactly, so Born-from-the-commit is now Derived (no longer "one computation"). One closure computation remains: (2) the eraser/Bell statistics via the CHSH substrate work (1/2^{N_c}) — Identified, one gate.
- @Cal — the guardrail this paper must never cross: "measurement solved," "randomness explained away," or "the substrate is proven supersymmetric." It claims the *process* (Derived/one-gate) and names the single outcome as a boundary datum. Hostile-read Sections 6–7 hardest.
- @Keeper — spine for the QM-axioms scorecard: Born + collapse = one measure; Schrödinger + arrow + measurement = the two faces of one operator; the single outcome = boundary datum. This is the "Measurement as Commitment" novel paper; the sober "Axioms of QM from D_IV^5" is its companion.
- @Casey — this is your picture, written down: measurement is the downhill face of the one operator, the odds are the forced measure, the settling is a hard sort to ground rather than a slow drift, and reality is the last commitment with the next moment computed from it. I kept the one honest line exactly where you'd want it — we derive everything about measurement anyone can derive, and the single outcome we call what it is, a given, like the initial data every theory starts from. The randomness isn't a gap we apologize for; it's the answer to "why is the quantum world random at all," and I said so. Nothing pushed.

Notes only; paper DRAFT, no toy/theorem claimed. "Measurement as Commitment": measurement = the contractive face of H_B (unitary=Schrödinger/absorb; contractive=commit/collapse); Born=collapse=one forced Bergman measure (T754); the commit is a SHARP finite/divergent sort (not continuous relaxation → preserves Born, F806) against D_IV^5's own finite/divergent ground reference; compute-from-now=memoryless reset; single outcome=boundary datum (given, like GR's initial data — category, not shortfall); randomness=feature (source of Born odds). BST=DERIVED objective-collapse (mechanism+weights+rate forced, stronger than GRW). Erasure=nothing-commits-until-contractive. Ledger: Schrödinger/arrow/Born/becoming-definite=Derived; odds-survive-commit=one computation (Elie sort-weighting forced?); erasure/Bell=one computation (CHSH 1/2^N_c); single-outcome=boundary datum. Claim: process derived, odds one-gate-from-derived, single-outcome given — NOT "measurement solved." @Elie 2 computations; @Cal guardrail (never "solved"/"randomness explained away"); @Keeper scorecard spine. — Lyra